

# FreeLoc: Online Floorplan Localization via Diffusion-Aided Pose Refinement

Haocheng Peng, Boyang Zhou, Jiarui Hu<sup>†</sup>, Xiyue Guo, Ziyang Zhang,  
Boming Zhao, Yifan Gao, Xiao Li, Hujun Bao, Zhaopeng Cui<sup>‡</sup>  
State Key Lab of CAD&CG, Zhejiang University

**Abstract:** Floorplans provide compact and widely available geometric maps for indoor localization, but existing high-performing floorplan-based methods still convert them into dense scene-specific offline databases, tying accuracy, storage, and runtime to the sampling resolution of the discretized pose space. We present FreeLoc, an online RGB-based floorplan localization framework that treats the floorplan as a directly queryable geometric map. FreeLoc introduces an efficient online geometric querying and diffusion-aided refinement scheme, which retrieves plausible pose anchors through on-the-fly floorplan ray querying and refines them into accurate continuous pose estimates. For sequential localization, FreeLoc develops an online likelihood construction strategy that bridges single-frame localization and probabilistic temporal fusion by constructing likelihoods from coarse-sampled candidates and refined pose hypotheses, enabling histogram-filter-based temporal fusion without offline databases. Experiments demonstrate real-time online inference and state-of-the-art performance in both single-frame and sequential localization, while real-world results validate practical deployability in indoor robotic localization scenarios.

**Keywords:** Floorplan Localization, Indoor Localization, Pose Refinement

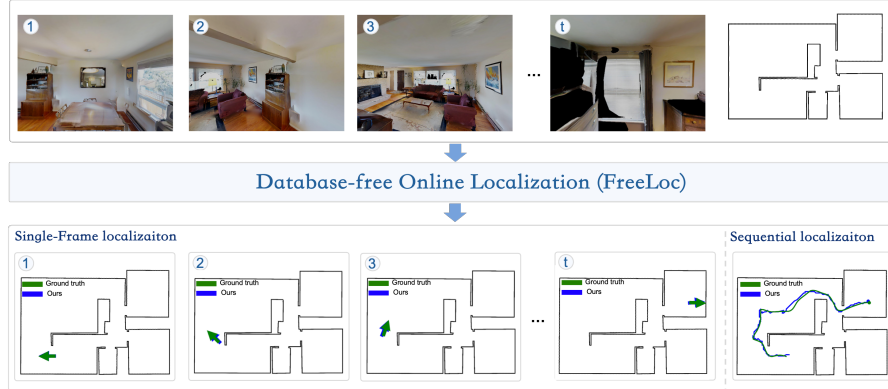


Figure 1: FreeLoc enables database-free online floorplan localization, directly using the provided floorplan to support accurate single-frame and sequential localization in new environments.

## 1 Introduction

Indoor camera localization is a core capability for robotic perception and embodied intelligence, providing the spatial grounding required for mobile robot navigation, human-robot interaction, and autonomous exploration. For robots operating in real indoor environments, a localization system

---

<sup>†</sup>Project leader

<sup>‡</sup>Corresponding author

should not only be accurate, but also be readily deployable in new scenes with minimal scene-specific preparation. Traditional indoor visual localization methods [1, 2, 3, 4] typically rely on pre-built image databases or accurate 3D scene models. These representations are costly to construct, storage-intensive, and difficult to maintain, making them difficult to scale across many buildings and adapt to changing indoor environments.

Floorplans offer a lightweight and scalable map representation for indoor localization. Compared with image databases and 3D scene models, they are compact, widely available [5, 6], and relatively stable over time, making them attractive for localization systems deployed across many buildings. Ideally, a floorplan should serve as a directly queryable geometric map, allowing localization without constructing additional scene-specific databases. However, existing high-performing floorplan localization methods [7, 8, 9] still largely rely on a database-driven formulation: they discretize the  $SE(2)$  pose space, precompute dense pose-indexed observations from the floorplan, and match query images against these entries for localization. In sequential settings, the same database-derived scores are further converted into observation likelihoods for histogram filtering, causing the database dependency to persist in temporal fusion. As a result, the limitation lies not merely in preprocessing cost, but in the discretized localization paradigm itself: once localization is tied to a finite pose-indexed database, localization precision, storage, and runtime all scale with the sampling resolution, creating an unfavorable trade-off among accuracy, efficiency, and scalability.

This motivates a more practical and challenging setting: *given only a floorplan, can we directly query it online to achieve accurate RGB-based single-frame and sequential localization in new environments, without constructing a scene-specific offline database?*

A useful clue comes from how humans localize themselves in unfamiliar indoor spaces. Rather than exhaustively comparing the current view with every possible location on the map, humans typically use the floorplan to narrow the search to a few plausible regions, and then refine their location by matching local geometric details with the current observation. However, realizing this coarse-to-fine intuition in an online localization system is non-trivial. First, the system must rapidly search the entire floorplan pose space online to identify plausible initial pose hypotheses. Second, these retrieved poses are inherently approximate: finite online sampling introduces discretization errors, while noisy and incomplete RGB-derived geometric observations further limit matching accuracy, making it challenging to recover accurate poses from such coarse hypotheses. Third, for sequential localization with histogram filtering, observation likelihoods over the discretized pose space are required to update the propagated motion prior, yet these likelihoods can no longer be densely evaluated from a precomputed scene-specific ray database.

To address these challenges, we propose FreeLoc, an online RGB-based floorplan localization framework that treats the floorplan as a directly queryable geometric map rather than a scene-specific offline retrieval database, as shown in Fig. 1. FreeLoc first retrieves a compact set of plausible pose anchors by querying geometric rays from the floorplan on the fly using a GPU-accelerated rotation-aware ray kernel. To refine these approximate anchors, we introduce a diffusion-aided pose refinement network that estimates residual pose corrections from the discrepancy between image-side and floorplan-side ray observations. Instead of relying on one-shot deterministic regression, we formulate pose refinement as an iterative denoising process, which better matches the nature of residual pose optimization and progressively corrects imperfect coarse anchors under ray-level geometric guidance. This allows FreeLoc to move beyond the resolution of coarse online samples and produce accurate pose hypotheses in continuous pose space without exhaustive dense sampling or precomputed scene-specific ray databases. To support sequential localization, we further integrate the online coarse-to-fine estimator into a probabilistic filtering pipeline. Rather than deriving dense observation likelihoods from an offline ray database, FreeLoc constructs observation likelihoods online from coarse-sampled candidates and refines their high-confidence regions using the refined pose hypotheses before Bayesian filtering [10, 11, 12]. This online likelihood construction bridges single-frame localization and temporal fusion without reintroducing a scene-specific database, enabling temporally consistent localization while preserving the immediate deployability of floorplans.

Our contributions are summarized as follows. First, we propose FreeLoc, an online coarse-to-fine framework for RGB-based floorplan localization that eliminates scene-specific offline database construction and enables immediate deployment in new environments. Second, we introduce an efficient online geometric querying and diffusion-aided refinement scheme, which retrieves plausible pose anchors directly from the floorplan and progressively refines them into accurate continuous pose estimates. Third, we develop an online likelihood construction strategy that bridges single-frame localization and probabilistic temporal fusion, enabling histogram-filter-based sequential localization by constructing observation likelihoods from online sampled candidates and updating them with refined pose hypotheses. Finally, experiments on various datasets demonstrate that FreeLoc achieves SOTA accuracy and efficiency across representative localization methods from different paradigms, while real-world experiments further validate its practicality for indoor robotic localization.

## 2 Related Work

**Visual Localization.** Visual localization is a fundamental problem in robotic perception and has been extensively studied for decades. Early methods commonly formulate it as image retrieval [1, 2, 13, 14, 15], where a query image is matched to a pre-built reference database to infer its pose. With advances in 3D vision, correspondence-based approaches [3, 4, 16, 17] construct 3D scene models, typically via structure-from-motion (SfM) [18], and estimate the 6-DoF camera pose from 2D-3D matches. Beyond ground-level reference maps, cross-view localization has also explored globally available satellite imagery as a spatial reference for ground-level localization [19]. More recently, neural scene representations such as NeRF [20] and 3DGS [21] have also been explored for pose estimation by relating query images to scene-specific representations [22, 23, 24]. However, these approaches depend on pre-constructed databases or scene-specific representations, leading to high storage costs and limited generalization to unseen environments.

**Floorplan Localization.** Floorplan localization estimates camera pose with respect to a given floorplan, offering a lightweight alternative to image databases and full 3D reconstructions. Early methods [25, 26, 27, 28] rely on LiDAR scans matched against floorplan geometry, while later approaches [29, 30, 31, 32] use depth cameras or visual odometry to obtain geometric observations. Although effective, these methods require dedicated sensors or assumptions such as known room or camera height, limiting practical applicability. To improve accessibility, recent works have explored RGB-only floorplan localization. LaLaLoc [33] and LaLaLoc++ [34] embed panoramic observations and floorplan-derived layouts into a shared latent space for retrieval. PF-Net [35] formulates localization as differentiable particle filtering, and LASER [36] aligns image features with rotation-aware circular map features rendered from rasterized floorplan geometry. However, many of these methods assume upright camera poses, making them less suitable for handheld or head-mounted settings. Another line of work [8, 36, 37, 38] incorporates semantic cues, but semantic annotations are often unavailable in standard floorplans. F<sup>3</sup>Loc [7] and UnLoc [9] take an important step toward practical RGB-based floorplan localization by using geometry-only floorplans and images captured under arbitrary camera poses. They predict floorplan depth from gravity-aligned images and localize the camera by matching the resulting rays against a dense pre-sampled floorplan-ray database. However, constructing such a database requires time-consuming scene-specific preprocessing and creates an inherent trade-off among localization accuracy, sampling density, and storage cost, limiting flexibility in new environments. In contrast, our method performs online localization without scene-specific offline ray databases, enabling efficient and accurate deployment in new scenes using only the provided geometric floorplan.

## 3 Method

In this work, we address RGB-based camera pose estimation in a known 2D floorplan. The key idea of FreeLoc is to treat the floorplan as an online geometric query interface rather than a precomputed pose-indexed database. Given a floorplan map, camera intrinsics  $\mathbf{K}$ , gravity direction, and either a single RGB image or an RGB sequence  $\mathcal{I} = \{I_t\}_{t=0}^T$  with inter-frame relative poses, our goal is to

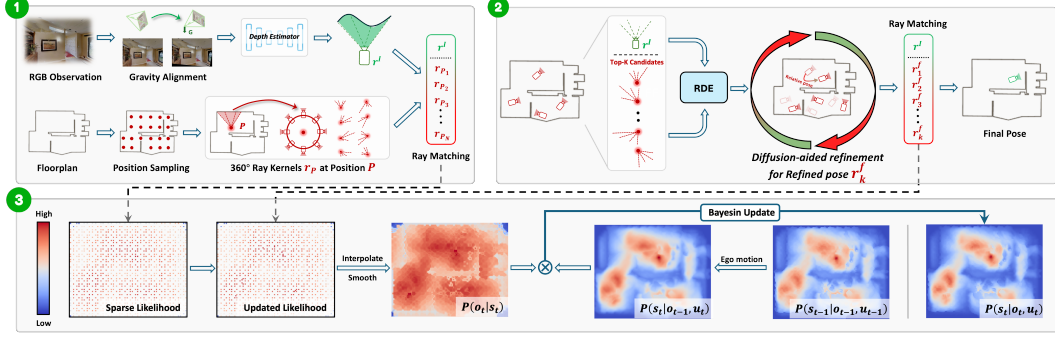


Figure 2: **Overview of FreeLoc.** We retrieve top- $K$  coarse anchors by matching image-side rays with online queried floorplan rays, then refine and re-score them for the final estimate. For sequential localization, coarse and refined candidates construct online likelihoods for temporal fusion.

estimate the camera pose  $\mathbf{s}_t = [s_{x,t}, s_{y,t}, s_{\phi,t}] \in \text{SE}(2)$  in the floorplan coordinate system, where  $(s_{x,t}, s_{y,t})$  denotes the 2D camera position and  $s_{\phi,t}$  denotes the yaw angle. As summarized in Fig. 2, FreeLoc builds a unified online coarse-to-fine framework: it retrieves pose anchors through on-the-fly floorplan ray querying, refines them with diffusion-aided residual optimization, and reuses the resulting hypotheses to construct online observation likelihoods. This enables both single-frame and sequential localization without scene-specific offline ray databases.

### 3.1 Single Frame Localization

**Online Coarse Pose Retrieval.** The first stage of FreeLoc retrieves a compact set of plausible pose anchors by querying floorplan geometry online, instead of matching the input image against a pre-computed pose-indexed ray database. This stage provides broad pose-space coverage for ambiguous indoor layouts while keeping the candidate set sufficiently sparse for efficient refinement.

Since RGB images in practical scenarios may be captured by handheld or head-mounted devices with arbitrary roll and pitch, we first normalize each image into a gravity-aligned view. Given the camera intrinsic matrix  $\mathbf{K}$  and the image roll  $r$  and pitch  $p$ , the rotation from the camera frame to the gravity-aligned frame is defined as  $\mathbf{R}_{gc} = (\mathbf{R}_z(r)\mathbf{R}_x(p))^\top$ . The roll and pitch can be obtained from inertial sensors available on modern mobile devices, or estimated by off-the-shelf single-image calibration methods such as GeoCalib [39] when sensor readings are unavailable. We then rectify the image with the homography  $\mathbf{H} = \mathbf{K}\mathbf{R}_{gc}\mathbf{K}^{-1}$ , which aligns the image vertical direction with gravity around the optical center. The rectified image, together with a visibility mask indicating invalid warped regions, is used to estimate floorplan depth and construct image-side ray observations.

Using the gravity-aligned image-mask pair, we follow F<sup>3</sup>Loc [7] to obtain image-side geometric observations by predicting floorplan depth and converting it into equiangular rays  $\hat{\mathbf{r}}_t^I$ . Unlike F<sup>3</sup>Loc that retrieves floorplan-side rays from a precomputed pose-indexed database, we query them online directly from the geometric floorplan. Specifically, we uniformly sample a compact set of pose hypotheses and evaluate their ray descriptors with a rotation-aware ray kernel. For each candidate pose  $(\mathbf{p}, \phi)$ , where  $\mathbf{p}$  is the 2D position and  $\phi$  is the yaw angle, the kernel casts  $V_r$  equiangular rays centered at  $\phi$  and computes the  $i$ -th ray distance by analytic boundary intersection:

$$d_i(\mathbf{p}, \phi) = \min_{b \in \mathcal{B}} \rho(\mathbf{p}, \phi + \Delta\theta_i, b), \quad (1)$$

where  $\mathcal{B}$  denotes the set of floorplan boundary segments,  $\Delta\theta_i$  is the angular offset of the  $i$ -th ray, and  $\rho(\cdot)$  returns the nearest valid ray-segment intersection distance.

Unlike existing database-based methods [7, 8, 9], which rely on dense hit-testing over rasterized occupancy grids, our query exploits the native line-segment structure of floorplans and computes analytic ray-boundary intersections directly. This reduces the query cost while preserving accurate geometric ray measurements. We implement the kernel as a vectorized batched GPU operation, allowing candidate poses and boundary segments to be evaluated in parallel. Together with compact pose sampling, this enables efficient online floorplan ray extraction for real-time coarse retrieval.

After obtaining floorplan-side rays, we rank sampled hypotheses by ray consistency:

$$\mathcal{P}_{t,j} = \exp \left( -\frac{\|\hat{\mathbf{r}}_{t,j}^{\mathcal{M}} - \hat{\mathbf{r}}_t^I\|_1}{\lambda} \right), \quad (2)$$

where  $\hat{\mathbf{r}}_{t,j}^{\mathcal{M}}$  denotes the scalar floorplan-side rays queried at the  $j$ -th candidate pose,  $\hat{\mathbf{r}}_t^I$  denotes the predicted image-side scalar rays, and  $\lambda$  controls the score sharpness. We retain the top- $K$  hypotheses for subsequent pose refinement as  $(\hat{\mathbf{r}}_t^I, \{(\mathbf{s}_{t,k}^c, \mathbf{r}_{t,k}^{\mathcal{M}})\}_{k \in \mathcal{K}_t})$ , where  $\mathcal{K}_t$  denotes the selected top- $K$  index set and  $\mathbf{s}_{t,k}^c$  is the  $k$ -th coarse pose anchor.

**Diffusion-Aided Pose Refinement.** After coarse retrieval, the top- $K$  anchors provide plausible but still discretized pose hypotheses. To move beyond the coarse sampling resolution, we propose a diffusion-aided pose refinement module that formulates refinement as residual pose optimization and produces accurate continuous pose estimates. For notational clarity, we omit the frame index  $t$  in this section. For the  $k$ -th coarse pose candidate  $\mathbf{s}_k^c \in \text{SE}(2)$ , we estimate a relative correction  $\Delta \mathbf{T}_k \in \text{SE}(2)$  conditioned on the image-side ray observation  $\hat{\mathbf{r}}^I$  and the corresponding floorplan-side ray observation  $\mathbf{r}_k^{\mathcal{M}}$ . The refined pose is obtained as

$$\mathbf{s}_k^f = \mathbf{s}_k^c \oplus \Delta \mathbf{T}_k, \quad (3)$$

where  $\oplus$  denotes pose composition in  $\text{SE}(2)$ . Inspired by recent optimization methods [40, 41, 42], we cast residual pose estimation as a conditional diffusion process rather than one-shot regression, as iterative denoising naturally mirrors pose optimization [40]. The residual correction is therefore progressively refined under the guidance of paired image-side and floorplan-side ray observations. We model the conditional residual distribution as  $p_\theta(\mathbf{x}_{0,k} \mid \hat{\mathbf{r}}^I, \mathbf{r}_k^{\mathcal{M}})$ , where  $\mathbf{x}_{0,k}$  parameterizes the residual pose in the local coordinate frame of  $\mathbf{s}_k^c$ . We encode the residual as  $\mathbf{x}_{0,k} = [\Delta x_k, \Delta y_k, \cos \Delta \phi_k, \sin \Delta \phi_k]^\top$ , where  $(\Delta x_k, \Delta y_k)$  denotes the residual translation and  $\Delta \phi_k$  denotes the residual yaw angle. The sine-cosine representation avoids discontinuities caused by angular wrap-around. During training, the ground-truth residual  $\mathbf{x}_{0,k}$  is computed from the relative transformation between the coarse pose  $\mathbf{s}_k^c$  and the ground-truth pose. We then apply the standard Gaussian forward diffusion process [43]:

$$\mathbf{x}_{\tau,k} = \sqrt{\bar{\alpha}_\tau} \mathbf{x}_{0,k} + \sqrt{1 - \bar{\alpha}_\tau} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (4)$$

where  $\tau$  denotes the diffusion timestep and  $\bar{\alpha}_\tau$  is the cumulative noise schedule. To condition the denoising process on the paired ray observations, we first encode the image-side and floorplan-side rays with a shared PointNet-style [44] ray encoder  $E_{\text{ray}}(\cdot)$ . We then introduce a Ray Discrepancy Encoder (RDE), denoted as  $\Phi(\cdot)$ , to compute soft correspondences between the paired ray features and aggregate them into a global conditioning feature:  $\mathbf{z}_k = \Phi(E_{\text{ray}}(\hat{\mathbf{r}}^I), E_{\text{ray}}(\mathbf{r}_k^{\mathcal{M}}), \hat{\mathbf{r}}^I, \mathbf{r}_k^{\mathcal{M}})$ , where  $\mathbf{z}_k$  encodes correspondence-aware geometric cues between the image-predicted rays and the floorplan rays. Further details of the RDE architecture are provided in Appendix A.2. The conditional denoiser  $D_\theta$  takes the noisy residual  $\mathbf{x}_{\tau,k}$ , the timestep  $\tau$ , and the conditioning feature  $\mathbf{z}_k$  as input, and is trained to predict the injected noise:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\tau, \mathbf{x}_{0,k}, \boldsymbol{\epsilon}} \left[ \|D_\theta(\mathbf{x}_{\tau,k}, \tau, \mathbf{z}_k) - \boldsymbol{\epsilon}\|_2^2 \right]. \quad (5)$$

At inference time, we initialize  $\mathbf{x}_{T,k}$  from pure noise and perform DDIM [45] sampling conditioned on  $\mathbf{z}_k$ . At each reverse denoising step, the residual is updated as

$$\hat{\mathbf{x}}_{\tau-1,k} = \sqrt{\bar{\alpha}_{\tau-1}} \left( \frac{\hat{\mathbf{x}}_{\tau,k} - \sqrt{1 - \bar{\alpha}_\tau} D_\theta}{\sqrt{\bar{\alpha}_\tau}} \right) + \sqrt{1 - \bar{\alpha}_{\tau-1}} D_\theta. \quad (6)$$

After iterating from  $\tau = T$  to  $\tau = 1$ , the denoised residual  $\hat{\mathbf{x}}_{0,k}$  is mapped to an  $\text{SE}(2)$  correction  $\Delta \hat{\mathbf{T}}_k$  and composed with the coarse hypothesis to obtain the refined pose in Eq. 3, yielding a locally consistent pose estimate guided by paired ray geometry.

**Training and Inference.** During training, we additionally sample a co-visible *gold anchor* near the ground truth to provide meaningful residual corrections and stabilize geometry-guided refinement.

At inference, no gold anchor is used; all top- $K$  anchors from coarse retrieval are refined in batch, re-queried online, and re-scored by Eq. 2 to select the final pose. In our experiments, we use a fixed budget of 2000 coarse sampled locations based on the average floorplan scale, with a rotation interval of  $15^\circ$ , and retain the top  $K = 10$  anchors for refinement. All experiments are conducted on the NVIDIA RTX 4090 GPU.

### 3.2 Sequential Localization

For robots with sequential RGB observations, sequential localization aims to maintain temporally consistent pose estimates. To bridge single-frame localization and temporal fusion, we construct observation likelihoods online from the sparse outputs of our coarse-to-fine estimator: coarse candidates provide broad pose-space support, while refined hypotheses sharpen high-confidence regions. The resulting likelihood is fused over time with a histogram filter [46].

Specifically, we first scatter the online coarse pose samples into the discretized SE(2) pose volume according to their matching scores, forming an initial sparse likelihood. The top- $K$  peak regions are then refined by our diffusion-aided pose refinement module, and their likelihood values are updated using the refined poses and confidence scores. The remaining sampled locations keep their initial coarse likelihoods. Finally, the updated sparse pose-score set  $\mathcal{A}_t$  is interpolated by  $\mathcal{I}(\cdot)$  and spatially smoothed by  $\mathcal{G}_\sigma$  to obtain  $p(o_t | s_t) = [\mathcal{G}_\sigma * \mathcal{I}(\mathcal{A}_t)](s_t)$ . This construction preserves the broad support of coarse retrieval while injecting more reliable likelihoods around refined high-confidence regions. Given the observation likelihood  $p(o_t | s_t)$ , we follow the histogram filtering formulation to perform sequential localization over the discretized pose space. The posterior distribution from the previous frame is first propagated by the inter-frame ego motion  $\mathbf{u}_t$ , yielding the prior distribution at the current frame:

$$p(s_t | o_{1:t-1}, \mathbf{u}_{1:t}) = \sum_{s_{t-1} \in \mathcal{S}} p(s_t | s_{t-1}, \mathbf{u}_t) p(s_{t-1} | o_{1:t-1}, \mathbf{u}_{1:t-1}). \quad (7)$$

The propagated distribution is then updated with the current observation likelihood to produce the posterior distribution at the current frame:

$$p(s_t | o_{1:t}, \mathbf{u}_{1:t}) = \eta p(o_t | s_t) p(s_t | o_{1:t-1}, \mathbf{u}_{1:t}), \quad (8)$$

where  $\eta$  is a normalization factor. We implement the transition model  $p(s_t | s_{t-1}, \mathbf{u}_t)$  with decoupled translation and rotation filters over the pose volume to accelerate the filtering update. At each time step, the final pose is obtained as the maximum a posteriori estimate.

## 4 Experiments

### 4.1 Experimental Setup

**Dataset.** We consider two widely used datasets with different distributions: Gibson [47] and Structured3D [48]. Gibson contains 118 indoor scenes with floorplans and monocular images captured with a  $108^\circ$  FoV, collected by F<sup>3</sup>Loc [7]. It includes Gibson(f), with 24,779 four-view sequences for single-frame localization, and Gibson(t), with 118 trajectories with 280 to 5,152 steps for sequential localization. Structured3D contains 3,296 furnished indoor environments and 78,453 perspective images captured with an  $80^\circ$  horizontal field of view. All evaluations follow official train/test splits.

**Baseline and Metrics.** We compare our method with representative floorplan localization approaches from different categories. F<sup>3</sup>Loc [7] is used as the primary baseline, as it is a strong state-of-the-art depth-based method, and we adopt its monocular version for comparison. We also evaluate against PF-Net [35] and LASER [36], two representative non-depth-based methods for differentiable particle-filter-based and feature-based floorplan localization, respectively. Following prior work [7, 36], for single-frame localization, we report recall at translation thresholds of 0.1m, 0.5m, and 1.0m, together with recall under the joint pose threshold of 1.0m and  $30^\circ$ . For sequential localization, we report success rate and tracking accuracy. A trajectory is considered successful at threshold  $X$  m if the predictions stay within  $X$  m of the ground truth over the last 10 frames. We

Table 1: Single-frame localization recall (%) on Gibson(f) and Structured3D. Bold denotes the best result.

| Method             | Gibson(f)        |                  |                |                   | Structured3D     |                  |                |                   |
|--------------------|------------------|------------------|----------------|-------------------|------------------|------------------|----------------|-------------------|
|                    | @0.1m $\uparrow$ | @0.5m $\uparrow$ | @1m $\uparrow$ | @1m30° $\uparrow$ | @0.1m $\uparrow$ | @0.5m $\uparrow$ | @1m $\uparrow$ | @1m30° $\uparrow$ |
| PF-Net             | 0                | 1.0              | 4.1            | 1.1               | 0.1              | 1.2              | 4.4            | 1.4               |
| LASER              | 0.2              | 3.9              | 9.8            | 6.2               | 0.5              | 5.2              | 10.0           | 7.8               |
| F <sup>3</sup> Loc | 5.3              | 30.7             | 37.9           | 36.3              | 1.7              | 15.0             | 23.0           | 21.9              |
| <b>Ours</b>        | <b>10.0</b>      | <b>39.4</b>      | <b>44.6</b>    | <b>42.9</b>       | <b>3.1</b>       | <b>23.9</b>      | <b>30.5</b>    | <b>29.1</b>       |

also report RMSE over the last 10 frames for both successful and all trajectories. Following F<sup>3</sup>Loc, our default sequential evaluation uses GT-derived inter-frame ego-motion for controlled comparison; robustness to noisy and temporarily unavailable ego-motion is evaluated in Sec. 4.5.

## 4.2 Single Frame Localization Results

**Results on Gibson(f) and Structured3D.** As shown in Table 1, our FreeLoc consistently outperforms all baselines on both Gibson(f) and Structured3D. Compared with the strongest baseline F<sup>3</sup>Loc, ours improves all recall metrics by 4.7–8.7 percentage points on Gibson(f) and 1.4–8.9 percentage points on Structured3D, demonstrating consistent gains across different data distributions. We attribute these improvements mainly to the proposed pose refinement stage. Unlike offline database matching, whose accuracy is constrained by the discretization of pre-sampled floorplan poses, our refinement module optimizes coarse anchors in a continuous pose space, allowing predictions to move beyond sampled locations and achieve finer pose accuracy. Moreover, while both F<sup>3</sup>Loc and our method benefit from floorplan-ray representations and substantially outperform PF-Net and LASER, FreeLoc further improves upon F<sup>3</sup>Loc by replacing dense offline ray matching with online coarse sampling and continuous pose refinement.

## 4.3 Sequential Localization Results

**Results on Gibson(t).** Table 2 reports the sequential localization results on Gibson(t), where the sequence length is set to 100 frames. Our method achieves the best success rate and overall tracking accuracy. Compared with F<sup>3</sup>Loc, FreeLoc improves the success rate by 13.5 percentage points and reduces RMSE<sub>all</sub> by 82.8%, demonstrating substantially more robust tracking over long trajectories. For RMSE<sub>succ</sub>, FreeLoc remains comparable to F<sup>3</sup>Loc, with only a marginal difference (0.15 m vs. 0.14 m), indicating that both methods achieve similarly low errors once successfully localized. This slight gap mainly stems from reconstructing dense observation likelihoods from sparse online candidates, which may slightly reduce local precision compared with dense likelihood evaluation over a precomputed ray database. Nevertheless, by initializing observation likelihoods from online sampled candidates and updating them with refined pose hypotheses, FreeLoc provides reliable frame-level evidence for histogram filtering, enabling stable sequential localization without an offline ray database. Detailed qualitative comparisons are provided in Appendix B.3.

Table 2: Sequential localization results on Gibson(t).

| Method             | SR@1m (%) $\uparrow$ | RMSE <sub>succ</sub> (m) $\downarrow$ | RMSE <sub>all</sub> (m) $\downarrow$ |
|--------------------|----------------------|---------------------------------------|--------------------------------------|
| PF-Net             | 8.0                  | 0.43                                  | 4.27                                 |
| LASER              | 35.0                 | 0.31                                  | 2.69                                 |
| F <sup>3</sup> Loc | 86.5                 | <b>0.14</b>                           | 0.87                                 |
| <b>Ours</b>        | <b>100.0</b>         | 0.15                                  | <b>0.15</b>                          |

**Timing.** We report the average per-frame inference time under the Gibson(t) sequential localization setting, together with the scene-specific offline sampling cost required by database-based methods. The offline sampling time is measured on the first Gibson(t) test scene, which covers approximately 10 m  $\times$  6 m. Following the official F<sup>3</sup>Loc implementation, we use its released sampling code and testing-time sampling resolution to construct the offline ray database. F<sup>3</sup>Loc achieves fast inference (0.014 s per frame) once the scene-specific ray database has been precomputed, but constructing this database takes 1.48 hours for a single scene. In contrast, our method localizes directly from the provided floorplan without offline ray-database construction, while still running about 15 Hz (0.069 s per frame) including online sampling. Therefore, when deployed in a new scene from

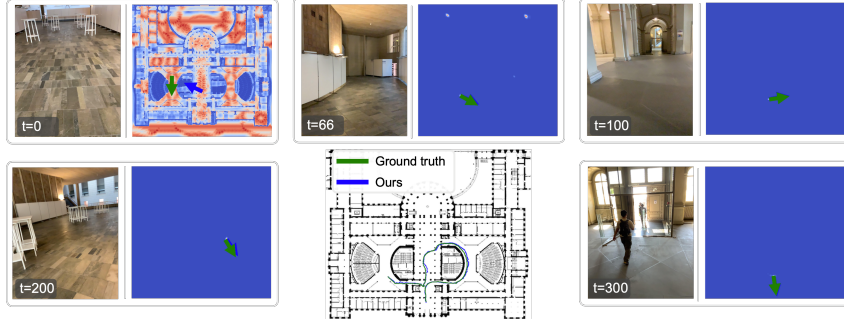


Figure 3: **Real-world performance of FreeLoc.** The center panel shows the floorplan with converged predicted and ground-truth trajectories; surrounding panels show time-ordered RGB observations and posterior probabilities. Blue indicates predictions and green indicates ground truth.

scratch, our unified coarse-to-fine framework substantially reduces the overall localization cost by eliminating the time-consuming scene-specific preprocessing stage.

#### 4.4 Real-World Experiment

To further validate real-world deployability, we build a real-world setting from the HGE building in the LaMAR dataset [49], since no public real-world indoor dataset provides both continuous RGB sequences and corresponding floorplan maps under our target setting. HGE is a large-scale indoor scene with rooms, corridors, and open halls, covering about  $130\text{ m} \times 130\text{ m}$ . The sequences are captured by a handheld iPhone with a narrow  $48^\circ$  FoV. As shown in Fig. 3, the posterior distribution is initially multi-modal due to the large search space and the ambiguity of narrow-FoV observations. FreeLoc gradually concentrates the posterior around the correct region and forms a stable, sharp peak within about 66 frames. After convergence, the estimated trajectory remains closely aligned with the ground truth throughout the sequence, demonstrating the stable long-term localization capability of FreeLoc in a large and structurally complex real-world environment. In terms of efficiency, FreeLoc enables faster scene preparation and deployment than  $F^3\text{Loc}$ , while still maintaining an inference speed of approximately 3 Hz on this large-scale indoor floorplan. Additional qualitative results are available on our project page: <https://zju3dv.github.io/freeloc/>.

#### 4.5 Ablation Studies

**Coarse-to-Fine Localization Analysis.** We analyze the behavior of *Diffusion-Aided Pose Refinement* under different qualities of coarse initialization. As shown in Table 3, when coarse hypotheses lie within 0.5 m, 1.0 m, and 1.5 m of the ground truth, diffusion refinement consistently reduces the average translation error, demonstrating effective residual pose correction from plausible initializations. In contrast, when all coarse hypotheses are considered regardless of their initialization quality, the improvement is marginal (3.14 m to 3.09 m). This limited gain arises mainly because the “All” set contains many poorly initialized hypotheses, including cross-room cases that are geometrically far from the ground-truth pose. Such hypotheses fall outside the effective range of local residual correction and are therefore difficult to recover through refinement alone. These results clarify the intended division of labor in our coarse-to-fine framework: *Online Coarse Pose Retrieval* narrows the global search to geometrically plausible pose hypotheses, while *Diffusion-Aided Pose Refinement* performs precise local residual correction to further improve localization accuracy.

Table 3: Pose refinement under different coarse-initialization quality.

| Initial error       | Before (m) | After (m) |
|---------------------|------------|-----------|
| $\leq 0.5\text{ m}$ | 0.25       | 0.18      |
| $\leq 1.0\text{ m}$ | 0.35       | 0.25      |
| $\leq 1.5\text{ m}$ | 0.42       | 0.32      |
| All                 | 3.14       | 3.09      |

**Diffusion-Aided Pose Refinement vs. ICP Optimization.** We further compare our diffusion-aided pose refinement with ICP optimization, a representative deterministic alternative for residual geometric alignment, to isolate the benefit of learned diffusion-based optimization. As shown in Table 4, diffusion refinement improves sequential SR@1m and SR@0.5m by 2.7 percentage points,

Table 4: Sequential localization comparison between ICP optimization and diffusion-aided pose refinement on Gibson(t).

| Refinement                      | SR@1m (%) $\uparrow$ | SR@0.5m (%) $\uparrow$ | SR@0.2m (%) $\uparrow$ | RMSE <sub>all</sub> (m) $\downarrow$ |
|---------------------------------|----------------------|------------------------|------------------------|--------------------------------------|
| ICP Optimization                | 97.3                 | 97.3                   | 43.2                   | 0.21                                 |
| Diffusion-Aided Pose Refinement | <b>100.0</b>         | <b>100.0</b>           | <b>51.4</b>            | <b>0.15</b>                          |

with a larger gain of 8.2 percentage points at the stricter 0.2 m threshold, while reducing RMSE<sub>all</sub> by 28.6%. These gains are achieved with only 6.8 ms additional latency (35.2 ms vs. 28.4 ms for ICP optimization), demonstrating that our learned diffusion-based optimizer provides more accurate pose refinement than deterministic ICP with only modest computational overhead.

**Robustness to Ego-Motion Uncertainty.** Following F<sup>3</sup>Loc, our default sequential evaluation uses GT-derived inter-frame ego-motion for controlled comparison. We further evaluate sequential localization robustness under three settings. The first two emulate common ego-motion errors: (i) frame-wise Gaussian noise models random motion-estimation errors, while (ii) motion-dependent noise increases with motion magnitude, with  $\sigma_t = b_t + a_t d_t$  and  $\sigma_r = b_r + a_r d_t + c_r d_r$ . The third setting models temporary ego-motion failure: (iii) ego-motion is unavailable for five consecutive frames, during which the last valid relative motion is reused under a constant-motion assumption. As shown in Table 5, FreeLoc maintains strong sequential performance under both noisy and temporarily unavailable ego-motion, demonstrating robust sequential localization under representative ego-motion disturbances.

Table 5: Robustness to inter-frame ego-motion uncertainty on Gibson(t).

| (a) Ego-motion robustness |          |       |                      |                     | (b) Noise parameters |                |       |      |      |
|---------------------------|----------|-------|----------------------|---------------------|----------------------|----------------|-------|------|------|
| Setting                   | Level    | SR@1m | RMSE <sub>succ</sub> | RMSE <sub>all</sub> | Noise                | Param.         | Mild  | Mod. | Sev. |
| Clean                     | –        | 100.0 | 0.145                | 0.145               | Gaussian             | $\sigma_t$ (m) | 0.01  | 0.03 | 0.05 |
|                           | Mild     | 100.0 | 0.147                | 0.147               |                      | $\sigma_r$ (°) | 0.5   | 1.0  | 2.0  |
| Gaussian                  | Moderate | 100.0 | 0.171                | 0.171               | Motion-dep.          | $b_t$ (m)      | 0.005 | 0.01 | 0.02 |
|                           | Severe   | 97.3  | 0.235                | 0.307               |                      | $a_t$          | 0.01  | 0.03 | 0.05 |
|                           | Mild     | 100.0 | 0.145                | 0.145               |                      | $b_r$ (°)      | 0.2   | 0.5  | 1.0  |
|                           | Moderate | 100.0 | 0.150                | 0.150               |                      | $a_r$ (°/m)    | 0.2   | 0.5  | 1.0  |
| Motion-dep.               | Severe   | 100.0 | 0.163                | 0.163               |                      | $c_r$          | 0.01  | 0.03 | 0.05 |
|                           | 5 frames | 100.0 | 0.158                | 0.158               |                      |                |       |      |      |

## 5 Conclusion

We presented FreeLoc, an online RGB-based floorplan localization framework that treats the floorplan as a directly queryable geometric map rather than a dense scene-specific offline database. FreeLoc builds a unified online coarse-to-fine pipeline that combines on-the-fly floorplan ray querying with diffusion-aided pose refinement to retrieve plausible anchors and refine them into accurate continuous pose estimates. The same coarse anchors and refined hypotheses are further used to construct online observation likelihoods for sequential localization, bridging single-frame localization and probabilistic temporal fusion without offline databases. Experiments on various datasets demonstrate accurate and efficient localization, as well as immediate deployability in new environments.

## 6 Limitations

Although our method demonstrates strong localization performance and favorable deployability, its generalization remains limited under substantially different camera settings or scene distributions. This limitation mainly stems from the monocular floorplan-depth estimator, which is sensitive to camera intrinsics, field of view, and scene scale. Moreover, while FreeLoc supports efficient inference in typical indoor environments, the enlarged search space of very large floorplans can still affect real-time performance, although FreeLoc retains faster and more convenient deployment than existing database-based approaches. Future work will explore more generalizable depth estimation and hierarchical floorplan querying to improve both cross-domain robustness and runtime scalability.

## Acknowledgments

We thank Yanqi Zhu for help with processing the Structured3D dataset. This work was partially supported by the NSFC (No. 62572425 and No. 624B2132).

## References

- [1] Y. Kalantidis, G. Toliás, Y. Avrithis, M. Phinikettos, E. Spyrou, P. Mylonas, and S. Kollias. Viral: Visual image retrieval and localization. *Multimedia Tools and Applications*, 51(2):555–592, 2011.
- [2] G. Berton and C. Masone. Megaloc: One retrieval to place them all. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 2886–2892, June 2025.
- [3] L. Liu, H. Li, and Y. Dai. Efficient global 2d-3d matching for camera localization in a large-scale 3d map. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 2372–2381, 2017.
- [4] P.-E. Sarlin, C. Cadena, R. Siegwart, and M. Dymczyk. From coarse to fine: Robust hierarchical localization at large scale. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 12716–12725, 2019.
- [5] A. Gueze, M. Ospici, D. Rohmer, and M.-P. Cani. Floor plan reconstruction from sparse views: Combining graph neural network with constrained diffusion. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, pages 1583–1592, October 2023.
- [6] R. H. Otten. Automatic floorplan design. In *19th Design Automation Conference*, pages 261–267. IEEE, 1982.
- [7] C. Chen, R. Wang, C. Vogel, and M. Pollefeys. F3loc: Fusion and filtering for floorplan localization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 18029–18038, 2024.
- [8] Y. Grader and H. Averbuch-Elor. Supercharging floorplan localization with semantic rays. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 27116–27125, 2025.
- [9] M. Wüest, F. Engelmann, O. Miksik, M. Pollefeys, and D. Barath. Unloc: Leveraging depth uncertainties for floorplan localization. *arXiv preprint arXiv:2509.11301*, 2025.
- [10] Z. Chen et al. Bayesian filtering: From kalman filters to particle filters, and beyond. *Statistics*, 182(1):1–69, 2003.
- [11] V. Fox, J. Hightower, L. Liao, D. Schulz, and G. Borriello. Bayesian filtering for location estimation. *IEEE pervasive computing*, 2(3):24–33, 2003.
- [12] F. Dellaert, D. Fox, W. Burgard, and S. Thrun. Monte carlo localization for mobile robots. In *Proceedings 1999 IEEE international conference on robotics and automation (Cat. No. 99CH36288C)*, volume 2, pages 1322–1328. IEEE, 1999.
- [13] R. Arandjelovic, P. Gronat, A. Torii, T. Pajdla, and J. Sivic. Netvlad: Cnn architecture for weakly supervised place recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 5297–5307, 2016.
- [14] O. Chum, J. Philbin, J. Sivic, M. Isard, and A. Zisserman. Total recall: Automatic query expansion with a generative feature model for object retrieval. In *2007 IEEE 11th international conference on computer vision*, pages 1–8. IEEE, 2007.

- [15] N. Keetha, A. Mishra, J. Karhade, K. M. Jatavallabhula, S. Scherer, M. Krishna, and S. Garg. Anyloc: Towards universal visual place recognition. *IEEE Robotics and Automation Letters*, 9 (2):1286–1293, 2023.
- [16] M. Kim, J. Koo, and G. Kim. Ep2p-loc: End-to-end 3d point to 2d pixel localization for large-scale visual localization. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 21527–21537, 2023.
- [17] P. Lindenberger, P.-E. Sarlin, V. Larsson, and M. Pollefeys. Pixel-perfect structure-from-motion with featuremetric refinement. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 5987–5997, October 2021.
- [18] O. Özyeşil, V. Voroninski, R. Basri, and A. Singer. A survey of structure from motion\*. *Acta Numerica*, 26:305–364, 2017.
- [19] X. Guo, H. Peng, J. Hu, H. Bao, and G. Zhang. From satellite to ground: Satellite assisted visual localization with cross-view semantic matching. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 3977–3983. IEEE, 2024.
- [20] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. *Communications of the ACM*, 65(1):99–106, 2021.
- [21] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4), July 2023. URL <https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/>.
- [22] B. Zhao, L. Yang, M. Mao, H. Bao, and Z. Cui. Pnerfloc: Visual localization with point-based neural radiance fields. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 7450–7459, 2024.
- [23] H. Zhai, X. Zhang, B. Zhao, H. Li, Y. He, Z. Cui, H. Bao, and G. Zhang. Splatloc: 3d gaussian splatting-based visual localization for augmented reality. *IEEE Transactions on Visualization and Computer Graphics*, 2025.
- [24] Z. Huang, H. Yu, Y. Shentu, J. Yuan, and G. Zhang. From sparse to dense: Camera relocalization with scene-specific detector from feature gaussian splatting. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 27059–27069, 2025.
- [25] F. Boniardi, T. Caselitz, R. Kümmerle, and W. Burgard. Robust lidar-based localization in architectural floor plans. In *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 3318–3324. IEEE, 2017.
- [26] X. Wang, R. J. Marcotte, and E. Olson. Glfp: Global localization from a floor plan. In *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 1627–1632. IEEE, 2019.
- [27] L. Gao and L. Kneip. Fp-loc: Lightweight and drift-free floor plan-assisted lidar localization. In *2022 International Conference on Robotics and Automation (ICRA)*, pages 4142–4148, 2022. doi:10.1109/ICRA46639.2022.9812361.
- [28] F. Xie and S. Schwertfeger. Robust lifelong indoor lidar localization using the area graph. *IEEE Robotics and Automation Letters*, 9(1):531–538, 2024. doi:10.1109/LRA.2023.3334158.
- [29] S. Ito, F. Endres, M. Kuderer, G. D. Tipaldi, C. Stachniss, and W. Burgard. W-rgb-d: floor-plan-based indoor global localization using a depth camera and wifi. In *2014 IEEE international conference on robotics and automation (ICRA)*, pages 417–422. IEEE, 2014.

- [30] H. Chu, D. K. Kim, and T. Chen. You are here: Mimicking the human thinking process in reading floor-plans. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, December 2015.
- [31] E. Wijmans and Y. Furukawa. Exploiting 2d floorplan for building-scale panorama rgb-d alignment. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, July 2017.
- [32] W. Winterhalter, F. Fleckenstein, B. Steder, L. Spinello, and W. Burgard. Accurate indoor localization for rgb-d smartphones and tablets given 2d floor plans. In *2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 3138–3143. IEEE, 2015.
- [33] H. Howard-Jenkins, J.-R. Ruiz-Sarmiento, and V. A. Prisacariu. Lalaloc: Latent layout localisation in dynamic, unvisited environments. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10107–10116, 2021.
- [34] H. Howard-Jenkins and V. A. Prisacariu. Lalaloc++: Global floor plan comprehension for layout localisation in unvisited environments. In *European Conference on Computer Vision*, pages 693–709. Springer, 2022.
- [35] P. Karkus, D. Hsu, and W. S. Lee. Particle filter networks with application to visual localization. In *Conference on robot learning*, pages 169–178. PMLR, 2018.
- [36] Z. Min, N. Khosravan, Z. Bessinger, M. Narayana, S. B. Kang, E. Dunn, and I. Boyadzhiev. Laser: Latent space rendering for 2d visual localization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11122–11131, 2022.
- [37] O. Mendez, S. Hadfield, N. Pugeault, and R. Bowden. Sedar: reading floorplans like a human—using deep learning to enable human-inspired localisation. *International Journal of Computer Vision*, 128(5):1286–1310, 2020.
- [38] N. Zimmerman, T. Guadagnino, X. Chen, J. Behley, and C. Stachniss. Long-term localization using semantic cues in floor plan maps. *IEEE Robotics and Automation Letters*, 8(1):176–183, 2022.
- [39] A. Veicht, P.-E. Sarlin, P. Lindenberger, and M. Pollefeys. Geocalib: Learning single-image calibration with geometric optimization. In *European Conference on Computer Vision*, pages 1–20. Springer, 2024.
- [40] J. Wang, C. Rupprecht, and D. Novotny. Posediffusion: Solving pose estimation via diffusion-aided bundle adjustment. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9773–9783, 2023.
- [41] W. Li, Y. Yang, S. Yu, G. Hu, C. Wen, M. Cheng, and C. Wang. Diffloc: Diffusion model for outdoor lidar localization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15045–15054, 2024.
- [42] J. Zhang, M. Wu, and H. Dong. Genpose: Generative category-level object pose estimation via diffusion models. *arXiv preprint arXiv:2306.10531*, 2023.
- [43] J. Ho, A. Jain, and P. Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.
- [44] C. R. Qi, H. Su, K. Mo, and L. J. Guibas. Pointnet: Deep learning on point sets for 3d classification and segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 652–660, 2017.
- [45] J. Song, C. Meng, and S. Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020.

- [46] S. Thrun. Probabilistic robotics. *Communications of the ACM*, 45(3):52–57, 2002.
- [47] B. Shen, F. Xia, C. Li, R. Martín-Martín, L. Fan, G. Wang, C. Pérez-D’Arpino, S. Buch, S. Srivastava, L. Tchapmi, et al. igibson 1.0: A simulation environment for interactive tasks in large realistic scenes. In *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 7520–7527. IEEE, 2021.
- [48] J. Zheng, J. Zhang, J. Li, R. Tang, S. Gao, and Z. Zhou. Structured3d: A large photo-realistic dataset for structured 3d modeling. In *European Conference on Computer Vision*, pages 519–535. Springer, 2020.
- [49] P.-E. Sarlin, M. Dusmanu, J. L. Schönberger, P. Speciale, L. Gruber, V. Larsson, O. Miksik, and M. Pollefeys. Lamar: Benchmarking localization and mapping for augmented reality. In *European Conference on Computer Vision*, pages 686–704. Springer, 2022.
- [50] I. Loshchilov and F. Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.
- [51] Å. Björck. Least squares methods. *Handbook of numerical analysis*, 1:465–652, 1990.
- [52] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao. Depth anything v2. *arXiv:2406.09414*, 2024.

## A Additional Implementation Details

### A.1 Training Details

For the floorplan-depth estimator, we follow F<sup>3</sup>Loc [7] and train the model for 100 epochs on both Gibson [47] and Structured3D [48] with a batch size of 32. For the diffusion-aided pose refinement network, we train the model for 200 epochs on each dataset with a batch size of 128. We use the AdamW optimizer [50] with an initial learning rate of  $1 \times 10^{-3}$ , together with a cosine annealing learning-rate scheduler. We adopt a cosine noise schedule during training and use 10-step DDIM sampling [45] at inference. The refinement network contains approximately 3.0M parameters.

### A.2 Ray Discrepancy Encoder Details

Given the image-side ray observation  $\hat{\mathbf{r}}^I$  and the floorplan-side ray observation  $\mathbf{r}_k^{\mathcal{M}}$  associated with the  $k$ -th coarse anchor, both represented as ray-indexed 2D point sets in  $\mathbb{R}^{V_r \times 2}$ , we first extract point-wise ray features using a shared PointNet-style [44] ray encoder:

$$\mathbf{F}^I = E_{\text{ray}}(\hat{\mathbf{r}}^I), \quad \mathbf{F}_k^{\mathcal{M}} = E_{\text{ray}}(\mathbf{r}_k^{\mathcal{M}}), \quad (9)$$

where  $\mathbf{F}^I, \mathbf{F}_k^{\mathcal{M}} \in \mathbb{R}^{V_r \times D}$ . The shared encoder maps the image-side and floorplan-side ray observations into a common feature space, enabling their local geometric patterns to be compared consistently. The Ray Discrepancy Encoder (RDE) is then applied on top of these point-wise features, together with the original ray coordinates, to construct a compact geometric condition for diffusion-aided pose refinement.

The goal of RDE is to capture how the image-side ray observation differs from the floorplan-side ray observation under the current coarse anchor. To establish flexible associations between the two ray sets, we first project their point-wise features into a normalized matching space:

$$\bar{\mathbf{F}}^I = \text{Normalize}(\Pi_I(\mathbf{F}^I)), \quad \bar{\mathbf{F}}_k^{\mathcal{M}} = \text{Normalize}(\Pi_{\mathcal{M}}(\mathbf{F}_k^{\mathcal{M}})), \quad (10)$$

where  $\Pi_I(\cdot)$  and  $\Pi_{\mathcal{M}}(\cdot)$  are learnable projection heads for the two ray features, and  $\text{Normalize}(\cdot)$  denotes  $\ell_2$  normalization along the feature dimension. We then compute the pairwise similarity matrix as

$$\mathbf{S}_k = \frac{\bar{\mathbf{F}}^I (\bar{\mathbf{F}}_k^{\mathcal{M}})^\top}{\gamma}, \quad (11)$$

where  $\gamma$  is a temperature parameter. Bidirectional soft correspondence matrices are obtained by applying row-wise softmax to  $\mathbf{S}_k$  and its transpose:

$$\mathbf{A}_k^{I \rightarrow \mathcal{M}} = \text{softmax}_{\text{row}}(\mathbf{S}_k), \quad \mathbf{A}_k^{\mathcal{M} \rightarrow I} = \text{softmax}_{\text{row}}(\mathbf{S}_k^\top). \quad (12)$$

This soft correspondence preserves the ray-wise structure while relaxing strict index-wise matching. Strict index-wise matching can be unreliable under pose misalignment or image-side ray prediction errors, since the ray indices between the two observations may become shifted and no longer form exact correspondences. By allowing each ray to attend to all candidate rays in the other observation, the proposed soft correspondence enables more flexible geometric matching and improves robustness to imperfect coarse anchors and noisy ray observations.

For the image-to-floorplan direction, we use  $\mathbf{A}_k^{I \rightarrow \mathcal{M}}$  to compute the softly matched floorplan-side feature and coordinate for each image-side ray:

$$\tilde{\mathbf{F}}_k^{\mathcal{M}} = \mathbf{A}_k^{I \rightarrow \mathcal{M}} \mathbf{F}_k^{\mathcal{M}}, \quad \tilde{\mathbf{r}}_k^{\mathcal{M}} = \mathbf{A}_k^{I \rightarrow \mathcal{M}} \mathbf{r}_k^{\mathcal{M}}. \quad (13)$$

We then construct a point-wise discrepancy descriptor by combining feature-level compatibility and ray-level geometric discrepancy:

$$\mathbf{Q}_k^{I \rightarrow \mathcal{M}} = \psi \left( [\mathbf{F}^I, \tilde{\mathbf{F}}_k^{\mathcal{M}}, \mathbf{F}^I - \tilde{\mathbf{F}}_k^{\mathcal{M}}, \mathbf{F}^I \odot \tilde{\mathbf{F}}_k^{\mathcal{M}}, \right. \\ \left. \hat{\mathbf{r}}^I, \tilde{\mathbf{r}}_k^{\mathcal{M}}, \hat{\mathbf{r}}^I - \tilde{\mathbf{r}}_k^{\mathcal{M}}, \hat{\mathbf{r}}^I \odot \tilde{\mathbf{r}}_k^{\mathcal{M}}] \right), \quad (14)$$

where  $\psi(\cdot)$  is a lightweight pair encoder,  $\odot$  denotes element-wise multiplication, and  $[\cdot]$  denotes concatenation along the feature dimension. The feature terms compare each source ray feature with its softly matched target feature, where the difference term highlights feature mismatch and the product term provides a multiplicative compatibility cue. The geometric terms perform the same comparison in the ray coordinate space, where the coordinate difference directly encodes the residual displacement between the paired ray observations, while the coordinate product provides an additional interaction cue. Together, these terms allow the descriptor to encode both compatibility in the learned geometric feature space and explicit discrepancy in the ray coordinate space. The floorplan-to-image descriptor  $\mathbf{Q}_k^{\mathcal{M} \rightarrow I}$  is constructed symmetrically by swapping the source and target ray observations.

Finally, RDE aggregates the point-wise discrepancy descriptors into a global ray discrepancy condition. Since the residual pose is determined by the overall discrepancy pattern between the two ray observations rather than by an isolated ray mismatch, we summarize the point-wise descriptors through complementary global pooling operations. A confidence head predicts a scalar importance score for each descriptor, which is normalized along the ray dimension:

$$\alpha_k^{I \rightarrow \mathcal{M}} = \text{softmax} \left( h(\mathbf{Q}_k^{I \rightarrow \mathcal{M}}) \right). \quad (15)$$

We combine confidence-weighted mean pooling with max pooling:

$$\mathbf{g}_{k,\text{mean}}^{I \rightarrow \mathcal{M}} = \sum_{i=1}^{V_r} \alpha_{k,i}^{I \rightarrow \mathcal{M}} \mathbf{Q}_{k,i}^{I \rightarrow \mathcal{M}}, \quad \mathbf{g}_{k,\text{max}}^{I \rightarrow \mathcal{M}} = \max_{i=1, \dots, V_r} \mathbf{Q}_{k,i}^{I \rightarrow \mathcal{M}}. \quad (16)$$

The confidence-weighted mean captures the dominant discrepancy trend across reliable ray correspondences, while max pooling preserves the most salient local mismatches that may indicate critical residual pose corrections. Together, they summarize both global consistency and local evidence in the ray discrepancy pattern, providing a compact condition for residual pose prediction. We perform the same aggregation for the floorplan-to-image direction and concatenate the two directional pooled features:

$$\begin{aligned} \mathbf{z}_k &= \Phi_{\text{RDE}} \left( \mathbf{F}^I, \mathbf{F}_k^{\mathcal{M}}, \hat{\mathbf{r}}^I, \mathbf{r}_k^{\mathcal{M}} \right) \\ &= [\mathbf{g}_{k,\text{mean}}^{I \rightarrow \mathcal{M}}, \mathbf{g}_{k,\text{max}}^{I \rightarrow \mathcal{M}}, \mathbf{g}_{k,\text{mean}}^{\mathcal{M} \rightarrow I}, \mathbf{g}_{k,\text{max}}^{\mathcal{M} \rightarrow I}]. \end{aligned} \quad (17)$$

The resulting feature  $\mathbf{z}_k$  compactly summarizes the bidirectional ray discrepancy under the  $k$ -th coarse anchor and serves as the geometric condition for the diffusion denoiser.

Overall, RDE is designed to extract residual pose cues from ray-level discrepancies. The soft correspondence mechanism improves robustness to imperfect coarse anchors and noisy ray observations, bidirectional matching captures complementary evidence from both image-to-floorplan and floorplan-to-image perspectives, and confidence-aware aggregation guides the denoiser to focus on reliable yet informative discrepancies for residual pose prediction.

### A.3 Real-World Dataset Details

Following F<sup>3</sup>Loc, we adapt LaMAR [49] to our real-world floorplan localization setting by constructing a subset with paired RGB observations, ground-truth poses, and a corresponding 2D floorplan. We use the iOS sequences from the HGE building. Since some trajectories extend beyond the building interior, we filter the data using the provided ground-truth poses and the spatial extent of the HGE building, retaining only indoor observations that fall within the HGE floorplan coverage, where floorplan-based localization is well-defined.

We obtain the complete HGE building floorplan from the official source and use it as the localization map. To align the original LaMAR annotations with our floorplan localization setting, we convert the provided SE(3) poses from the global world coordinate system into SE(2) poses in the floorplan coordinate system. Specifically, we manually select four salient landmarks that can be reliably identified in both the RGB observations and the floorplan. For each landmark, we record its 2D position in the LaMAR world coordinate system and its corresponding pixel coordinate on the floorplan. These four point correspondences provide eight constraints, which are sufficient to



Figure 4: **Qualitative results of diffusion-aided pose refinement.** Starting from coarse pose anchors retrieved by online floorplan sampling, our diffusion-aided pose refinement module progressively predicts residual pose corrections guided by paired image-side and floorplan-side ray geometry. The color transition from light blue to dark blue indicates the pose recovery process, while green denotes the ground truth. Coarse anchors from different initial locations are refined toward poses closer to the ground truth, validating the effectiveness of diffusion-aided pose refinement.

estimate the six parameters of a 2D affine transformation in a least-squares [51] manner. We then apply the estimated affine transformation to all ground-truth trajectory positions. For orientation, we extract the yaw heading from the original quaternion and transform it consistently into the floorplan coordinate system using the linear part of the estimated affine transformation. This yields the final SE(2) pose annotations on the floorplan.

To better match the floorplan geometry to the visual observations, we manually remove non-geometric annotations such as room numbers and stair labels. We also remove doors at traversable connections along the collected trajectories while preserving other doors as solid walls, since they are mostly closed during LaMAR data collection. Landmark selection, floorplan alignment, and preprocessing use no validation or test trajectories and are performed solely to adapt LaMAR to the floorplan-localization setting, rather than to tune the localization method. After preprocessing, we split the retained data into training, validation, and test sets. Quantitative evaluations of all methods on this real-world setting are reported in Sec. C.1.

## B Additional Qualitative Results

### B.1 Diffusion-Aided Pose Refinement Results

Fig. 4 presents additional qualitative results of our diffusion-aided pose refinement module. The results show that FreeLoc consistently refines coarse pose anchors toward the ground-truth poses across different scenes and locations. Even when the initial coarse samples are located at varying distances from the target pose, the refinement module effectively corrects their residual translation and orientation errors, producing refined hypotheses that are closer to the ground truth. These qualitative results demonstrate that the proposed module can move beyond discretized coarse samples and perform accurate continuous pose optimization.

### B.2 Single-Frame Localization Results

Fig. 5 presents qualitative single-frame localization results on Gibson and Structured3D. The results show that FreeLoc produces more accurate pose estimates than PF-Net [35], LASER [36], and F<sup>3</sup>Loc [7] across different scenes. Without relying on a scene-specific offline ray database, Free-

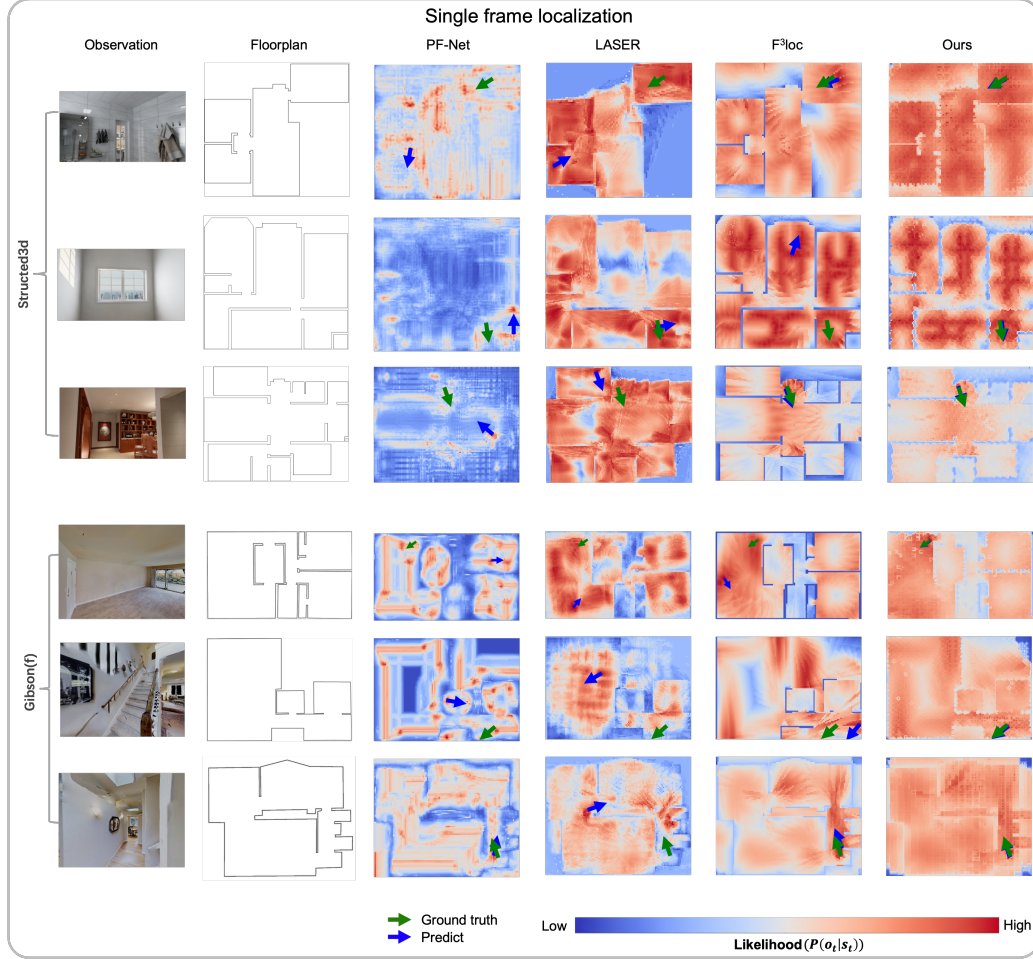


Figure 5: **Qualitative comparison of single-frame localization results.** We compare FreeLoc with representative baselines on Gibson and Structured3D. Compared with PF-Net, LASER, and F<sup>3</sup>Loc, FreeLoc produces pose estimates that are better aligned with the ground truth, benefiting from accurate online coarse pose retrieval and diffusion-aided pose refinement.

Loc achieves better single-frame localization than F<sup>3</sup>Loc by retrieving reliable coarse pose anchors online and further refining them in continuous pose space through diffusion-aided pose refinement.

### B.3 Sequential Localization Results

Fig. 6 presents qualitative sequential localization results across different scenes. The results show that FreeLoc converges faster and maintains more accurate localization than the baselines. By constructing observation likelihoods online from reliable coarse anchors and diffusion-refined candidates, FreeLoc provides more accurate frame-level evidence for temporal filtering, allowing the posterior distribution to concentrate around the correct region more quickly and remain stable over time. Unlike F<sup>3</sup>Loc, which relies on a scene-specific offline ray database, our online likelihood construction improves deployment flexibility in new environments.

## C Additional Quantitative Results

### C.1 Real-World Results

Table 6 presents the quantitative results on the real-world HGE building dataset. The HGE building is a challenging floorplan localization setting, covering a large spatial extent of approximately

Table 6: Real-world quantitative localization results on the HGE building. – indicates that no trajectory was successfully localized.

| Method             | Single-frame Localization |                  |                |                   | Sequential Localization |                                       |                                      |
|--------------------|---------------------------|------------------|----------------|-------------------|-------------------------|---------------------------------------|--------------------------------------|
|                    | @0.1m $\uparrow$          | @0.5m $\uparrow$ | @1m $\uparrow$ | @1m30° $\uparrow$ | SR@1m (%) $\uparrow$    | RMSE <sub>succ</sub> (m) $\downarrow$ | RMSE <sub>all</sub> (m) $\downarrow$ |
| PF-Net             | 0                         | 0                | 0              | 0                 | 0                       | –                                     | 76.9                                 |
| LASER              | 0                         | 0.3              | 0.7            | 0.7               | 0                       | –                                     | 41.4                                 |
| F <sup>3</sup> Loc | 0                         | 0                | <b>1.7</b>     | <b>1.7</b>        | <b>66.7</b>             | <b>0.2</b>                            | 15.1                                 |
| Ours               | <b>0.3</b>                | <b>1.3</b>       | <b>1.7</b>     | <b>1.7</b>        | <b>66.7</b>             | 0.4                                   | <b>14.5</b>                          |

130 m  $\times$  130 m with complex indoor layouts and repetitive structural patterns. As a result, all methods achieve relatively low single-frame recall in this real-world scenario. Moreover, both F<sup>3</sup>Loc and our method rely on image-side floorplan-depth prediction. In such a large-scale real-world environment, ray depths can span tens of meters in many regions, which substantially increases depth-estimation uncertainty and limits the overall reliability of single-frame localization. Despite this difficulty, depth-based methods can still produce relatively concentrated single-frame likelihoods in regions with moderate ray depths and sufficient geometric constraints. These likelihoods provide useful frame-level evidence for temporal filtering, enabling sequential localization to reach a reliable level of accuracy over time.

For single-frame localization, our method achieves the best performance under the stricter thresholds, especially at 0.1 m and 0.5 m. This demonstrates the benefit of diffusion-aided pose refinement, which optimizes coarse anchors in the continuous pose space and produces more accurate pose hypotheses around plausible locations.

For sequential localization, the concentrated single-frame likelihoods are accumulated through temporal filtering, substantially improving localization stability. Our method achieves the same SR@1m as F<sup>3</sup>Loc while reducing RMSE<sub>all</sub> from 15.1 m to 14.5 m. The RMSE<sub>succ</sub> of our method is slightly higher than F<sup>3</sup>Loc (0.4 m vs. 0.2 m). This small gap mainly comes from reconstructing dense observation likelihoods by interpolating a compact set of online-retrieved and refined pose-score candidates, whereas F<sup>3</sup>Loc evaluates likelihoods over a dense offline ray database. Such interpolation may slightly reduce local precision for already successful trajectories, but it enables database-free online localization while maintaining competitive sequential accuracy in real-world deployment.

## C.2 Ablation Study of Diffusion-Aided Pose Refinement

We first compare our diffusion-aided pose refinement with ICP optimization in the main paper. Since residual pose refinement can naturally be formulated as a geometric registration problem, ICP serves as a representative deterministic alternative. However, ICP relies on reliable geometric correspondences and local alignment, whereas our ray observations are sparse and imperfect due to coarse pose errors and image-side ray prediction noise, limiting the effectiveness of deterministic alignment. This comparison demonstrates the benefit of learned refinement over conventional geometric optimization, but does not isolate whether the gain originates from the learned ray-discrepancy representation or the diffusion formulation itself. We therefore further introduce a simpler learned direct-regression baseline to specifically examine the contribution of diffusion-based refinement.

For a controlled comparison, both learned variants use the same RDE to construct the global ray-discrepancy feature, with matched network capacity and identical training settings. The only difference lies in the refinement formulation: the direct-regression variant replaces iterative diffusion denoising with a one-shot regression head that directly predicts the relative pose correction.

As shown in Table 7, diffusion-aided refinement consistently outperforms direct regression across all reported metrics, including a 2.8 percentage-point gain in SR@0.2m and a 25% reduction in RMSE<sub>all</sub>. Since the feature representation, model capacity, and training configuration are controlled, these improvements directly isolate the benefit of the diffusion-based refinement formulation. At

Table 7: Ablation study of diffusion-aided pose refinement against direct regression on Gibson(t).

| Refinement                      | SR@1m (%) $\uparrow$ | SR@0.5m (%) $\uparrow$ | SR@0.2m (%) $\uparrow$ | RMSE <sub>all</sub> (m) $\downarrow$ |
|---------------------------------|----------------------|------------------------|------------------------|--------------------------------------|
| Direct Regression               | 97.3                 | 97.3                   | 48.6                   | 0.20                                 |
| Diffusion-Aided Pose Refinement | <b>100.0</b>         | <b>100.0</b>           | <b>51.4</b>            | <b>0.15</b>                          |

the same time, the competitive performance of direct regression indicates that the RDE provides informative geometric conditioning.

Compared with one-shot regression, iterative denoising is better suited to residual pose optimization by progressively refining pose corrections toward geometrically consistent solutions. Moreover, noisy and locally ambiguous ray-level discrepancies make a single deterministic prediction less flexible in capturing uncertainty in the residual correction. By modeling the residual correction as a conditional distribution and progressively denoising it under geometric guidance, diffusion-aided refinement produces more accurate pose hypotheses. Together, the ICP and direct-regression comparisons show that the performance gains arise not only from learned geometric representations, but also from the diffusion-based refinement formulation itself.

### C.3 Compatibility with Stronger Depth Estimation

We further compare with UnLoc [9], which extends F<sup>3</sup>Loc with stronger foundation-model-based monocular depth estimation using Depth Anything V2 [52]. Since depth quality directly affects the image-side geometric observations, we use the same UnLoc depth predictions for FreeLoc to enable a controlled comparison of the localization pipelines. Specifically, we replace our default depth predictions with those from UnLoc at test time while keeping the FreeLoc pipeline and diffusion refinement model unchanged, without retraining.

As shown in Table 8, FreeLoc consistently outperforms UnLoc across all single-frame metrics on both Gibson(f) and Structured3D, with gains of 0.6–3.1 percentage points and 0.4–1.8 percentage points, respectively. In sequential localization, FreeLoc further improves the success rate from 97.3% to 100.0% and reduces RMSE<sub>all</sub> from 0.28 m to 0.15 m. Since both methods use the same depth predictions, these results demonstrate that FreeLoc maintains superior localization accuracy under stronger depth estimation. Moreover, directly adopting UnLoc depth without retraining demonstrates the plug-and-play compatibility of our pipeline with stronger monocular depth estimators.

Table 8: Controlled comparison with UnLoc under matched depth predictions. FreeLoc uses the same UnLoc depth predictions at test time without retraining the diffusion refinement model. Bold denotes the better result.

| (a) Single-frame localization |                  |                  |                |                   |                  |                  |                |                   |
|-------------------------------|------------------|------------------|----------------|-------------------|------------------|------------------|----------------|-------------------|
| Method                        | Gibson(f)        |                  |                |                   | Structured3D     |                  |                |                   |
|                               | @0.1m $\uparrow$ | @0.5m $\uparrow$ | @1m $\uparrow$ | @1m30° $\uparrow$ | @0.1m $\uparrow$ | @0.5m $\uparrow$ | @1m $\uparrow$ | @1m30° $\uparrow$ |
| UnLoc                         | 13.5             | 46.1             | 49.6           | 47.8              | 3.6              | 27.7             | 33.6           | 32.6              |
| <b>Ours</b>                   | <b>16.6</b>      | <b>46.9</b>      | <b>50.2</b>    | <b>48.7</b>       | <b>4.9</b>       | <b>29.5</b>      | <b>34.3</b>    | <b>33.0</b>       |

| (b) Sequential localization on Gibson(t) |                      |                                       |                                      |
|------------------------------------------|----------------------|---------------------------------------|--------------------------------------|
| Method                                   | SR@1m (%) $\uparrow$ | RMSE <sub>succ</sub> (m) $\downarrow$ | RMSE <sub>all</sub> (m) $\downarrow$ |
| UnLoc                                    | 97.3                 | 0.16                                  | 0.28                                 |
| <b>Ours</b>                              | <b>100.0</b>         | <b>0.15</b>                           | <b>0.15</b>                          |

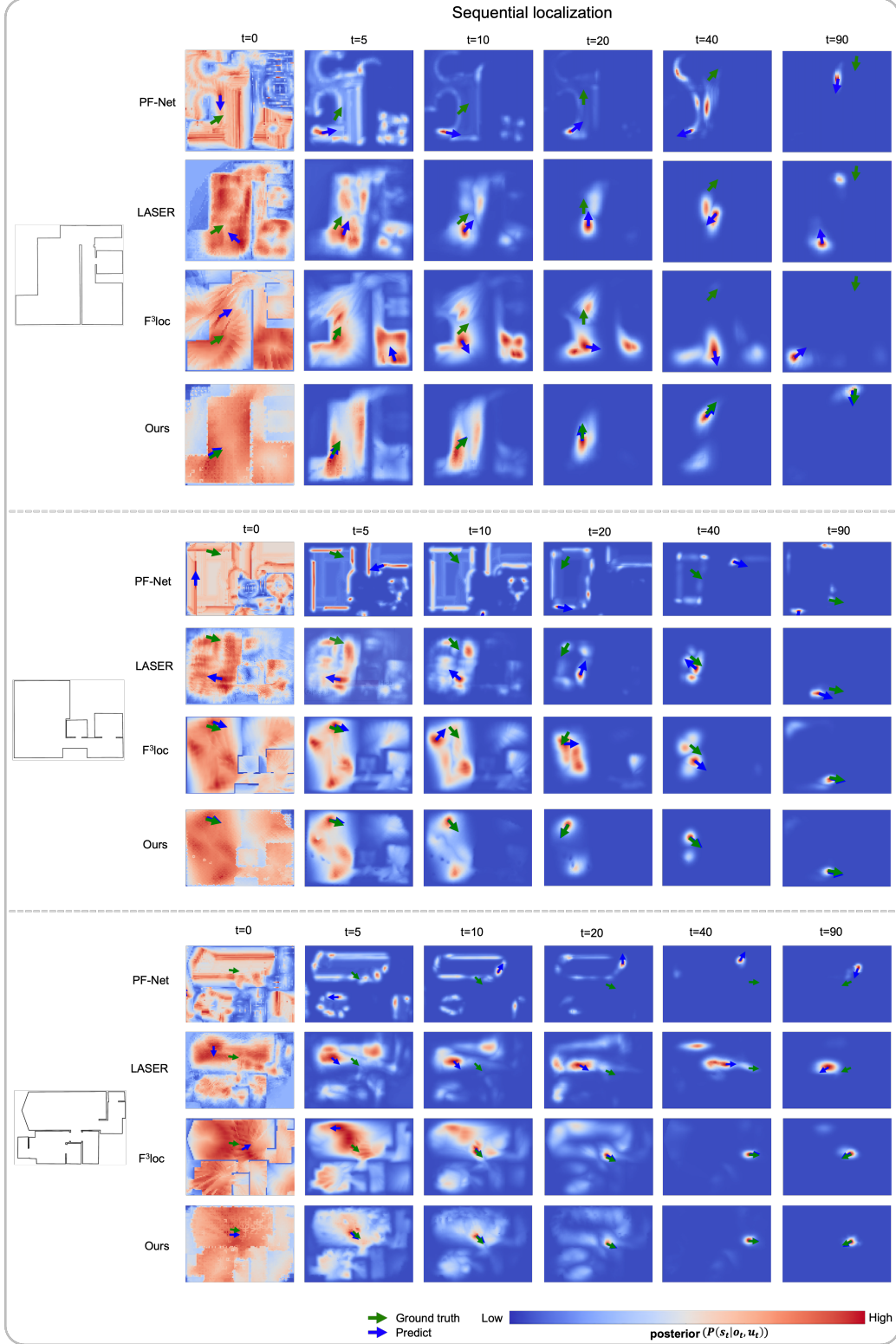


Figure 6: **Qualitative comparison of sequential localization results.** We compare FreeLoc with representative baselines on the Gibson(t) sequential localization dataset. The results show the temporal evolution of posterior distributions. Benefiting from more reliable single-frame localization evidence, FreeLoc concentrates the posterior around the correct region more quickly and maintains more accurate localization over time.